

Karim Almomani

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Computer Engineering student interested in embedded systems and firmware development, with experience in ARM Cortex-M microcontrollers, embedded C, and the integration of hardware and software.

EDUCATION

McMaster University

Hamilton, Canada

B.Eng. in Computer Engineering, GPA: 3.80/4.00

EXPERIENCE

EcoCAR McMaster – EV Design Competition Team

Hamilton, Canada

Systems Design & Integration Team Member

Oct 2025 – Present

- High Voltage subteam member on an EV design competition team, which came in second place in 2026
- Co-presented a PCB design workshop in Altium Designer
- Performed hands-on inspection of automotive inverters and High Voltage Cable maintenance
- Documented and carried out High Voltage safety procedures
- Performed soldering tasks on PCBs for the Vehicle Inspection subteam

Orange Telecommunications

Amman, Jordan

Software Development Intern

May 2025 – Aug 2025

- Developed responsive web pages integrating APIs, HTML5, CSS, Bootstrap, and JavaScript
- Mentored incoming interns by setting onboarding goals and providing ongoing technical feedback

PROJECTS

3D Environment Scanner

2DX4: MCU Course Final Project

Winter 2026

- Engineered Embedded C firmware on an ARM Cortex-M4 (TI MSP432E401Y) to rotate a VL53L1X Time-of-Flight sensor via I2C and a GPIO-controlled stepper motor
- Implemented polling for sensing button inputs (scan/pause/return), & UART (115200 baud) sending CSV data to the host PC
- Built a Python program (PySerial, Matplotlib) to parse serial data, flag anomalies, and create a 3D render of surroundings

Digital Logic Design Labs

2DI4: Logic Design

Fall 2025

- Designed and implemented finite state machines on FPGA boards using Intel Quartus
- Designed simple RAM modules as part of laboratory and coursework

EV Charger Finder

Sole Developer

Summer 2025

- Developed a program allowing users to find EV chargers which are compatible with their cars, within a selected city
- Integrated APIs and the leaflet.js map library

3-Bit Digital-To-Analogue Converter

2EI4: Bonus Electronics Project

Winter 2026

- Simulated (LTSpice) and designed a 3-bit binary-weighted-resistor DAC with an inverting summing op-amps, achieving 0.714 V/step over 0-5V, while limiting input current draw to <10 nA by utilizing buffer and comparator op-amps.
- Tested performance using Digilent Analog Discovery 2's oscilloscope, measuring -1% gain error

ZenLenZ (Phone Stabilization System)

MedSprint Hackathon

Design Lead

March 2025

- Designed a joystick-controlled camera stabilizer to aid users suffering from ALS in taking pictures with their phones
- Took complete responsibility for the CAD and 3D printed it, acting as the frame and axles for the device

Snake Game

2SH4: C++ Course Project

Fall 2025

- Developed a CLI Snake game in C++ using pointers and dynamic memory allocation
- Implemented core game logic, such as input handling and memory management

SKILLS

Programming: C/C++, Python, JavaScript

Embedded Systems: Embedded C, ARM Cortex-M, UART, I2C

Tools: Keil uVision, LTSpice, Altium, Intel Quartus, Digilent WaveForms, RealTerm, Git, GitHub, Autodesk Inventor (CAD)